

ChE 427 (Entrepreneurship in Chemical Engineering)
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University of Illinois at Chicago
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Anderson Quiz:

- 1- Motivation - The most important thing for an entrepreneur is making money.
Is to be their own Boss
- 2- Mission - Potential employees would rather work for a company that has a noble mission than one that pays the highest salary.
Not always. Some people will want to work because of the salary while others because the company mission align with their values and purpose
- 3- Intellectual property - I have written a computer code. Should I protect my intellectual property with a trademark, patent, or copyright?
Copyright
- 4- Team Building - It is better to find co-founders who think alike, rather than think differently.
Not always, sometimes you need to have someone who thinks differently
- 5- Strategic planning - Deciding whether to have a sale next week is an example of strategic thinking.
It is not necessarily an example of strategic thinking because it can be a part of a larger strategic plan, depending on the thought process behind the decision. ChatGPT
- 6- Sales/Marketing
 - If a product is good enough and marketed well enough, salespeople aren't necessary.
Not necessarily true depending on the market
 - The most important task of the marketing department is writing good ads and maintaining the website.
Not exactly. Those are tactics, but the real core of marketing is understanding the customer and market. ChatGPT
 - Product/Market Fit - An amazing product idea with limited sales is better than a boring product with massive sales.
The reverse is often true. Investors and markets value traction (sales, users, adoption) over just "amazing ideas." A "boring" product that solves a real pain point and sells consistently (e.g., payroll software, plumbing supplies) often beats a flashy product with no buyers. Product/Market Fit means customers want what you're building so badly that sales almost "pull themselves. Without that, even the best idea fails. ChatGPT

- 7- Funding – Good investors squeeze as much money from founders to increase their return on investment.

Good investors want founders motivated and supported, as the startup's success depends heavily on the founders.

- 8- Accounting – What is the profit of a company with \$1,000,000 in annual revenue, fixed costs of \$100,000 per year, and variable costs of \$400,000 per year? What is the percent profit margin?

*Percent profit margin = (profit/Revenue) * 100*

Total Cost = \$100,000 + \$400,000=\$500,000

Revenue = \$1,000,000

Profit=\$1,000,000-\$500,000=\$500,000

*Percent profit margin = (\$500,000/\$1,000,000) * 100=50%*

- 9- Finance – My company can earn \$200,000 profit per year by investing \$3,000,000. Should I borrow money for that project with an interest rate of 8%?

Invest \$3,000,000 → earn \$200,000 per year (≈6.7% return). Only borrow if expected return > cost of capital. ChatGPT

- 10- Cash flow – our company buys parts from overseas with upfront payment and 9-10 weeks lead time. It takes our production team 4 weeks to build and ship products to customers. Our COGs are 50% and labor cost is 15% of selling price, what payment terms should we request from our customers to stay cash flow positive.

Require 50% deposit at order and 50% on shipment, or full prepayment. Net terms (e.g., Net 30) extend your cash flow deficit. ChatGPT

- 11- Manufacturing – Manufacturing work force is abundant, and most manufacturing workers do not need math or computer skills.

Modern manufacturing (automation, robotics, quality control, lean processes) increasingly requires technical, math, and computer literacy. Abundant workforce ≠ trained workforce. Upskilling is key. ChatGPT

- 12- Human Resources – Objective is to hire people at the lowest possible salary

It should be to hire at a fair, competitive salary that attracts and retains talent.

- 13- Economics – A good way to increase profit is to raise the price

A good way to increase profit is to optimize price relative to value → sometimes raising, sometimes lowering, sometimes bundling. ChatGPT

- 14- Legal – Pick one – You need a lawyer to help you find ways around laws that restrict your business. Your lawyer needs to know the law. (You can provide the business

background). A good lawyer keeps you out of trouble, not waiting until you have messed up.

A good lawyer knows the law, keeps you compliant, and prevents costly mistakes.

15- Global business – The primary reason globalization is important is that it gets us a source of inexpensive labor.

It is not only about cheap labor but It's about access to new markets, resources, ideas, technology, and supply chains, as well as labor advantages. ChatGPT

16- Project management – To make sure a project gets finished on time you should plan for extra workers and overtime during the last month.

good project management = detailed upfront planning, milestones, and resource allocation throughout. Prevention > last-minute scrambling. ChatGPT

17- Environment – If you don't pollute either air or water, you are good to go.

Environmental responsibility also includes waste management, energy use, carbon emissions, biodiversity impact, and regulatory compliance. ChatGPT

18- Customer service –

- Providing good customer service is very costly.

Not necessarily. It often reduces costs (fewer returns, fewer complaints, higher efficiency). ChatGPT

- Providing good customer service improves profit

True. Loyal, satisfied customers buy more, refer others, and reduce churn → driving profitability. ChatGPT

19- Research and development – Ideas for new products usually come from PhD scientists.

Not true. most product ideas come from customers, market needs, frontline employees, and even competitors. ChatGPT

20- Ethics – True or False – Safe design of bridges is the responsibility of the manager of engineering.

True: Managers are legally and ethically responsible for ensuring safety standards are met ChatGPT